1A

```
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: 2d8cf058e153cc2ad6c95354c17359a6458658de1e7d206b19acd84af97826cf
Object stored at: .pes/objects/2d8cf058e153cc2ad6c95354c17359a6458658de1e7d206b19acd84af97826cf
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B

```
tanuj@error666:~/os-u4-orange-problem$ find .pes/objects -type f
.pes/objects/2d8cf058e153cc2ad6c95354c17359a6458658de1e7d206b19acd84af97826cf
.pes/objects/75d14791b254e2d944fd8e762c2317d59ff534e59fbc109771fe8f9ecb76c1ce
.pes/objects/d307fceb8cec6e7170f1edd75007c3b3d1917321931412914124d68b23aec04f
tanuj@error666:~/os-u4-orange-problem$
```

2B

```
tanuj@error666:~/os-u4-orange-problem$ make test_tree
./test_tree
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
tanuj@error666:~/os-u4-orange-problem$
```

2B

```
tanuj@error666:~/os-u4-orange-problem$ xxd .pes/objects/26b25d457597a7b0463f9620f666dd10aa2c4373a505967c7c8d70922a2d6ece
 | head -20
00000000: 0200 0000 0000 0000 0000 0000            ............
tanuj@error666:~/os-u4-orange-problem$
```

3A

```
tanuj@error666:~/os-u4-orange-problem$ ./pes init
echo "hello again" > file2.txt
./pes add file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  object.c
  untracked:  pes.h.save
  untracked:  Makefile
  untracked:  test_sequence.sh
  untracked:  test_tree
  untracked:  index.h
  untracked:  README.md
  untracked:  commit.c
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  tree.h
  untracked:  commit.h
  untracked:  index.c
  untracked:  pes.h
  untracked:  tree.c
  untracked:  pes.c
  untracked:  .gitignore
```

3B

```
tanuj@error666:~/os-u4-orange-problem$ cat .pes/index
304444 e8d5dc5f3f4d8cf90a4cf7f5fd2bdc34b5e076621f012aa3800aef88d0b69b04 1776577014 24 file1.txt
304444 48690a4a8fc73f13e590a45eae0b9ff39bd6e4b004ca3ae6205d3a267a9b231d 1776577079 12 file2.txt
tanuj@error666:~/os-u4-orange-problem$
```

4A

```
tanuj@error666:~/os-u4-orange-problem$ ./pes log
commit b3bf3b4d19f48f69f6b4763c73f98954559c8e6e41657e6d3488a65fe3297bde
Author: Tanuj PES2UG24CS553
Date:   1776577014

    third commit

commit 190485750ac3812f69c2018a88226f807e82d53662aa00c90a6e9104a047219a
Author: Tanuj PES2UG24CS553
Date:   1776577014

    second commit

commit c932920159322596b7ec2ca8060e185a15310789a488ba6a4f1d7526c7c35101
Author: Tanuj PES2UG24CS553
Date:   1776576918

    first commit
```

4B

```
tanuj@error666:~/os-u4-orange-problem$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/190485750ac3812f69c2018a88226f807e82d53662aa00c90a6e9104a047219a
.pes/objects/26b25d457597a7b0463f9620f666dd10aa2c4373a505967c7c8d70922a2d6ece
.pes/objects/273144f152e9d3c92afd2da82444fc45be86924a60abb8fefd071b6958e71fc9
.pes/objects/48690a4a8fc73f13e590a45eae0b9ff39bd6e4b004ca3ae6205d3a267a9b231d
.pes/objects/b3bf3b4d19f48f69f6b4763c73f98954559c8e6e41657e6d3488a65fe3297bde
.pes/objects/c932920159322596b7ec2ca8060e185a15310789a488ba6a4f1d7526c7c35101
.pes/objects/c9f1c61b06b80ddee5a4223ba8a91f2f8985ff4a3d58cf53975928b1a6563e29
.pes/objects/e8d5dc5f3f4d8cf90a4cf7f5fd2bdc34b5e076621f012aa3800aef88d0b69b04
.pes/refs/heads/main
tanuj@error666:~/os-u4-orange-problem$
```

4C

```
tanuj@error666:~/os-u4-orange-problem$ cat .pes/refs/heads/main
cat .pes/HEAD
b3bf3b4d19f48f69f6b4763c73f98954559c8e6e41657e6d3488a65fe3297bde
ref: refs/heads/main
tanuj@error666:~/os-u4-orange-problem$
```

```
tanuj@error666:~/os-u4-orange-problem$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 1d3608641419... Initial commit

Log after first commit:
commit 1d3608641419f55195f478be737e2b32c256c15e14dc6483d9bc50b00edad655
```

```
Author: Tanuj PES2UG24CS553
Date:   1776577291

    Initial commit


--- Second Commit ---
Committed: cf2d9435cd45... Update file.txt

--- Third Commit ---
Committed: 49a99b119b56... Add farewell

--- Full History ---
commit 49a99b119b561d24b29b8d3fe6f226e1d93896a3d604188acd0934a841b5ff8b
Author: Tanuj PES2UG24CS553
Date:   1776577291

    Add farewell

commit cf2d9435cd45a6eb5d042a2d52c36ccfe8eed7b38a42bb977caf09c219e93118
Author: Tanuj PES2UG24CS553
Date:   1776577291

    Update file.txt

commit 1d3608641419f55195f478be737e2b32c256c15e14dc6483d9bc50b00edad655
Author: Tanuj PES2UG24CS553
Date:   1776577291
```

```
    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
49a99b119b561d24b29b8d3fe6f226e1d93896a3d604188acd0934a841b5ff8b

--- Object Store ---
Objects created:
8
.pes/objects/1d3608641419f55195f478be737e2b32c256c15e14dc6483d9bc50b00edad655
.pes/objects/26b25d457597a7b0463f9620f666dd10aa2c4373a505967c7c8d70922a2d6ece
.pes/objects/273144f152e9d3c92afd2da82444fc45be86924a60abb8fefd071b6958e71fc9
.pes/objects/49a99b119b561d24b29b8d3fe6f226e1d93896a3d604188acd0934a841b5ff8b
.pes/objects/b36f2cb9757714ab0106b8bc2be30e6bb00104b70c7ee7b0ba02aee6fe85a39c
.pes/objects/c3865fb6ab55658ddb19241bbce2adf653471a8b28e39ef39d53fa4545e3590a
.pes/objects/cf2d9435cd45a6eb5d042a2d52c36ccfe8eed7b38a42bb977caf09c219e93118
.pes/objects/fa8c0fda0055999229a580e6f04fbb08bee7b58f3a311fbccb395998a675e647

=== All integration tests completed ===
```

Q5.1
To implement pes checkout <branch>, read the branch file in .pes/refs/heads/ to get the commit hash. Update .pes/HEAD to point to that branch, then restore the working directory files using that commit's tree. This is complex because modified files, deleted files, and nested folders must be handled safely.

Q5.2
Compare working directory files with the index using size, modification time, or hash. If a tracked file has local changes and the same file is different in the target branch, checkout must stop. This prevents overwriting uncommitted work.

Q5.3
Detached HEAD means HEAD points directly to a commit instead of a branch. New commits can still be made, but no branch tracks them. To recover them, create a new branch pointing to that commit hash.

Q6.1
Start from all branch heads and mark every reachable commit, parent commit, tree, and blob. Then scan .pes/objects/ and delete anything not marked reachable. A hash set is best for storing reachable hashes. For 100,000 commits and 50 branches, several hundred thousand objects may be visited.

Q6.2
GC is dangerous during commit because new objects may be created before the branch reference is updated. GC may think those objects are unused and delete them, causing corruption. Git avoids this using locks, atomic updates, and delayed cleanup.